

Spending Across the 2010's

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Background:

Understanding spending behavior is important for businesses to remain competitive within the economic market. Analyzing when customers spend the most, their preferred transaction methods, and how spending varies across age groups can provide insights into financial trends and decision-making patterns. Following the financial recession, the 2010s provided insight into a period of economic recovery, during which customer spending began recovering. At the same time, advancements in technology led to a significant increase in online transactions and digital payment methods. These changes reshaped how customers spend and manage their finances. By studying transaction data across this period, this project aims to better understand seasonal spending patterns, long-term trends, and demographic differences in financial behavior among age groups.

Outline:

Our course project uses the Financial Transactions Dataset: Analytics (2009 - 2019) from Kaggle: <https://www.kaggle.com/datasets/computingvictor/transactions-fraud-datasets>

The goal of our project was to extract data to visualize and understand changes in consumer spending behaviours over the course of a year and during periods of economic instability. We chose this dataset because it includes anonymized financial transaction records over a 10-year period, including details about whether a transaction was fraudulent. The dataset components consist of transaction data, card information, merchant category codes, fraud labels, and user data.

The key features we focused on were:

- Transaction amount
- Transaction date
- Transaction method (swipe, chip, online)
- User age
- Credit limit

This dataset is well suited for analyzing financial trends because it covers the economic recovery following the recession in 2009, along with the greatest period of growth for online transactions during the 2010s.

Major Questions:

For this project, we extracted data to analyze and answer the following questions:

- What times of year are busiest for commerce?
- How does spending change in the years following a recession (2010-2019)?
- How does the volume of online spending change as the internet matures?
- How does spending change across age groups on average, and who spends the highest percentage of their available credit?

Tools:

For our project, we used PostgreSQL, Python, Tableau, and Docker.

- PostgreSQL - Used to store and manage our cleaned data, allowing for efficient queries across the dataset.
- Python - Used to clean our data, load the database, and perform some EDA analysis.
- Tableau - Used for our final report and visualization. Paired with our use of Python and PostgreSQL, Tableau made it easy to visualize and analyze our data further after filtering and cleaning.
- Docker - Used to create and run a reproducible PostgreSQL environment for the project.

Data Pipeline:

- Data acquisition/loading: Downloaded the dataset from Kaggle and prepared the raw files for use in the project.
- ETL / cleaning: Removed formatting issues, standardized values, converted data types, and cleaned fields before loading the data into PostgreSQL. This included handling missing values and invalid placeholders where necessary.
- Storage of cleaned data: Stored cleaned data in PostgreSQL.
- Exploratory Data Analysis (EDA): Used summary statistics, trend-based views, and grouped comparisons to examine transaction patterns over time, transaction methods, and age groups.
- Reporting: Created visuals and dashboards in Tableau to present the final results.

Ethics and Considerations:

Since our project uses anonymized transaction data, there are no direct privacy concerns for vendors or customers in any given transaction. Some ethical concern could be given to the usage of metrics extracted from this data, as exploiting consumer spending habits could be considered unethical. This project is purely analytical and offers no advice for exploiting consumer practices.

The source dataset is licensed under the Apache 2.0 conditions, meaning we are free to use it as long as the source is properly credited and our use remains consistent with the license terms.

Results and Discussion:

What times of year are busiest for commerce? (Figure 1)

To find the busiest times of year for commerce, we examined monthly spending patterns across each year in the dataset, shown in Figure 1. The results show a fairly consistent monthly pattern each year, with February consistently appearing as the weakest month for spending. Spending increases and peaks in March, May, July, October, and December, with December appearing to be one of the strongest months overall. This suggests that commerce follows recurring monthly trends rather than changing randomly throughout the year. This can suggest that spending is positively correlated with consumer holidays like Christmas and summer vacations

How does spending change in the years following a recession (2010-2019)? (Figure 1)

To see how spending changed in the years following the recession, we examined the total spending volume over time from 2010 to 2019, shown in Figure 1. The results suggest that spending generally increased in the years following the recession, particularly from 2010-2014. After 2015, spending appeared to level off at a higher overall level. This supports the idea that consumer spending recovered over time following the recession and then became more stable later in the 2010s.

How does the volume of online spending change as the internet matures? (Figure 1)

To explore how online spending changed as the internet matured, we examined online transaction's share of volume across the decade, shown in Figure 1. The results show a clear upward trend in online transaction amounts from the early 2010s to around 2015. After 2015, the online transaction amount begins to level off, remaining higher than it was at the beginning of the time period. The percentage of spending done online shifted from around 14 percent to 16 percent with a peak of around 17 percent. This suggests that online transactions became a more

important part of overall consumer spending as digital commerce continued to grow.

What times of year are busiest for commerce?

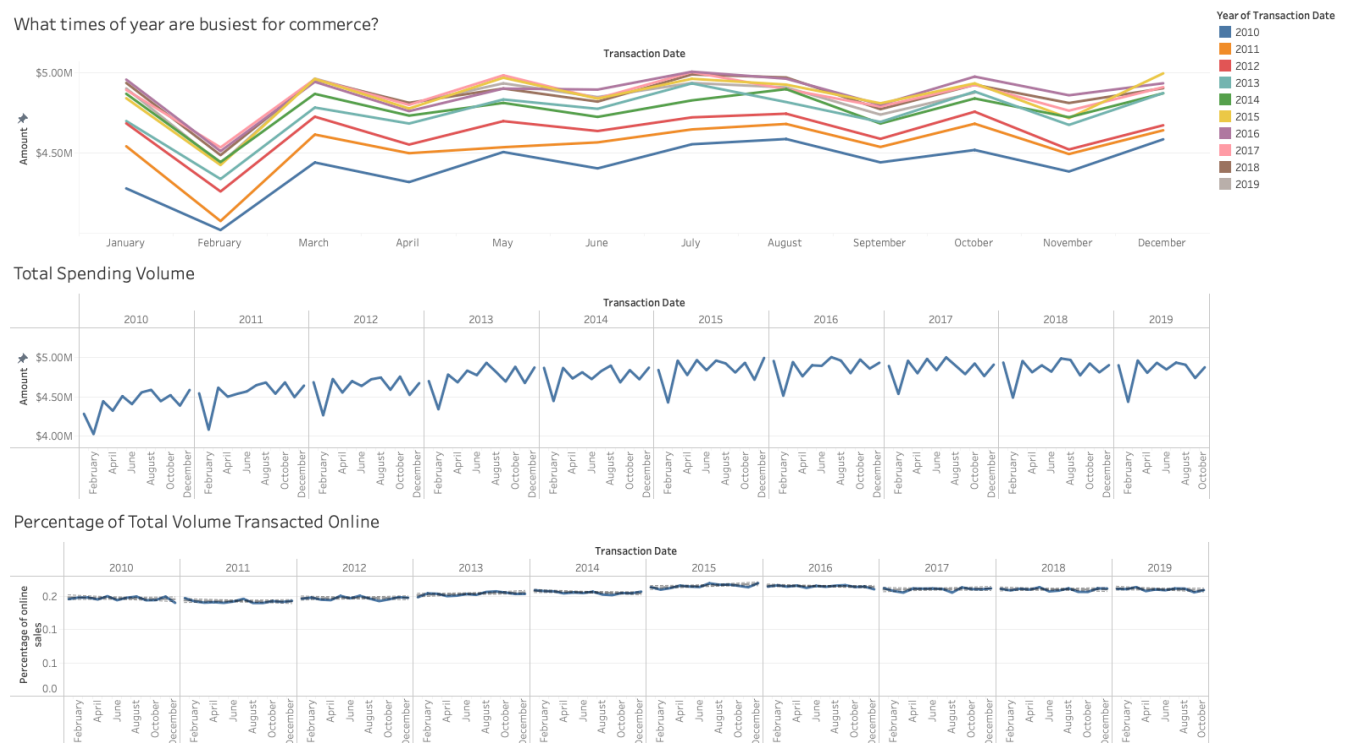


Figure 1. Monthly and yearly spending trends, including seasonal transaction patterns, total spending volume, and online transaction growth from 2010 to 2019.

How does spending change across age groups on average, and who spends the highest percentage of their available credit? (Figure 2)

To find how spending changed across age groups and which group used the highest percentage of available credit, we compared both spending per client and credit usage by age group, shown in Figure 2. The results suggest that older age groups, especially ages 45 to 54 and 55+, generally had the highest spending per client. In contrast, the 18 to 24 age group showed the highest credit usage percentage, often by a large margin compared to the other age groups. This suggests that while older individuals may spend more overall, younger people tend to use more of their available credit. Suggesting that banks were more apprehensive to lend to younger individuals following the 2008 banking reforms. The data is slightly skewed as the financial data does not contain equal length histories for 18-24 year olds, resulting in the group having less spending data than older groups. This can be seen as the sharp rise in spending during 2014-2016 when the group was beginning to have credit cards and begin spending.

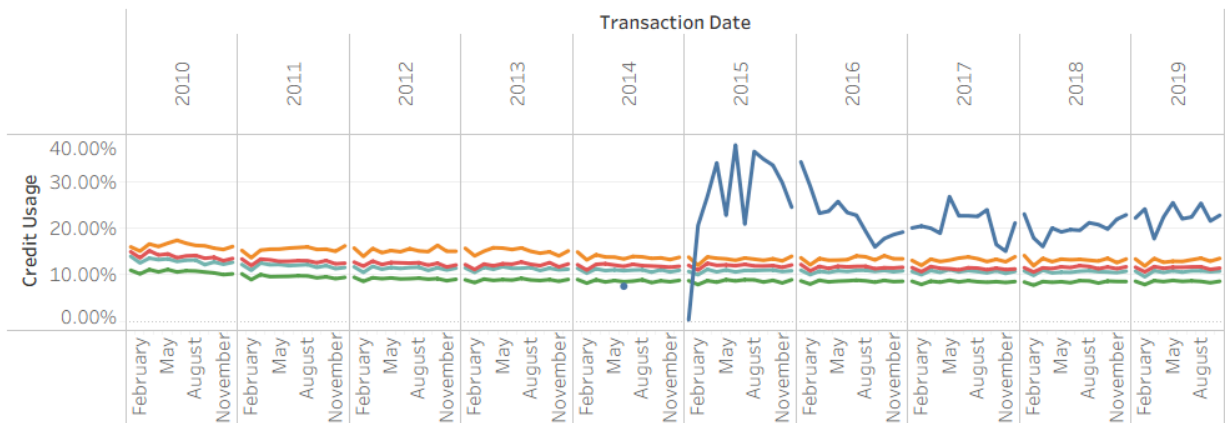
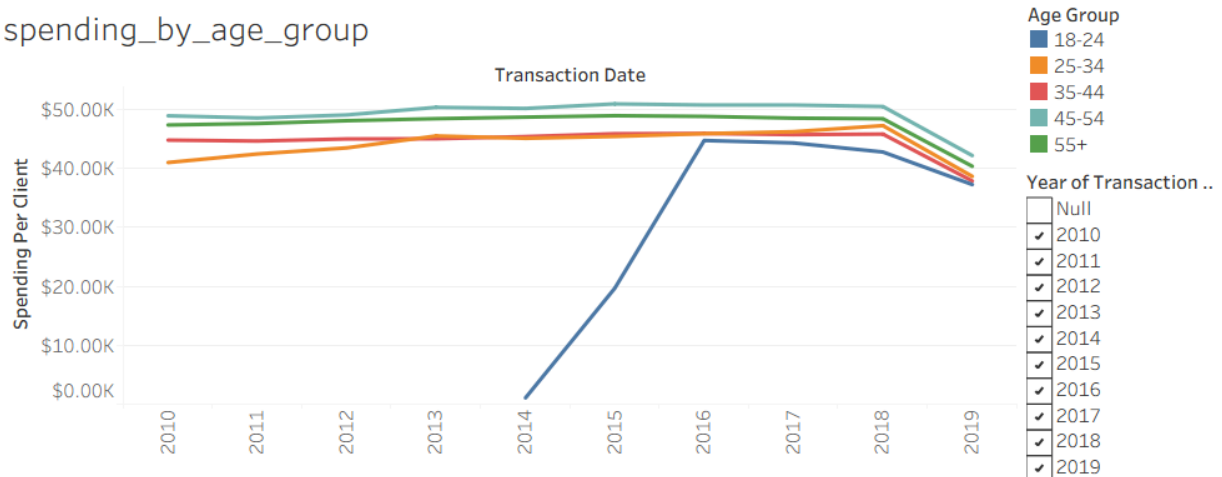


Figure 2. Comparison of spending per client and credit usage across age groups over time.

Exploratory Data Analysis.

Most of our EDA is performed by our [eda.py](#) script in the datapipeline.

Transactions missingness:

Total Rows	Date Missing	Amount Missing	Chip Missing	Merchant ID Missing	Merchant State Missing	MCC Missing
13,305,915	0	0	0	0	1,563,700	0

Clients missingness:

Total Rows	Age Missing	Gender Missing	Credit Score Missing	Income Missing	Debt Missing
2,000	0	0	0	0	0

Cards missingness:

Total Rows	Credit Limit Missing	Card Type Missing	Card Brand Missing	Has Chip Missing
6,146	0	0	0	0

Duplicate ids:

Table Name	Duplicate IDs
clients_data	0
cards_data	0
transactions_data	0

Transaction amount summary stats:

Total	Min	P25	Median	P75	Max	Average	Std Dev
13,305,915	-\$500.00	\$8.93	\$28.99	\$63.71	\$6,820.20	\$42.98	\$81.66

Monthly transaction volume:

Unavailable in Report due to size, refer to Figure 3

Transactions by method:

Use Method	Transaction Count	Avg Amount
Swipe Transaction	6,967,185	\$41.21
Chip Transaction	4,780,818	\$40.89
Online Transaction	1,557,912	\$57.26

Average transaction amounts by age group:

Age Group	Transaction Count	Avg Amount	Total Amount
18-24	16,221	\$47.72	\$774,023.00
25-34	1,229,086	\$47.08	\$57,861,650.00
35-44	2,808,998	\$40.79	\$114,570,300.00
45-54	3,517,726	\$43.49	\$152,973,400.00
55-64	2,613,263	\$42.54	\$111,160,800.00
65+	3,120,621	\$43.10	\$134,495,400.00

Correlation: amount vs credit limit

Metric	Correlation Value
Amount vs. Credit Limit	0.069374

Additionally, Figure 3 provides an exploratory summary of the dataset through a heatmap of transaction counts by month and year, yearly household spending distributions, credit availability over time, total spending per client, and a transaction location map. This exploratory view helped provide context for the broader trends examined in the later figures and spikes in December and Summer months supported the decision to focus on seasonality, long-term spending growth, online transactions, and age-group differences.

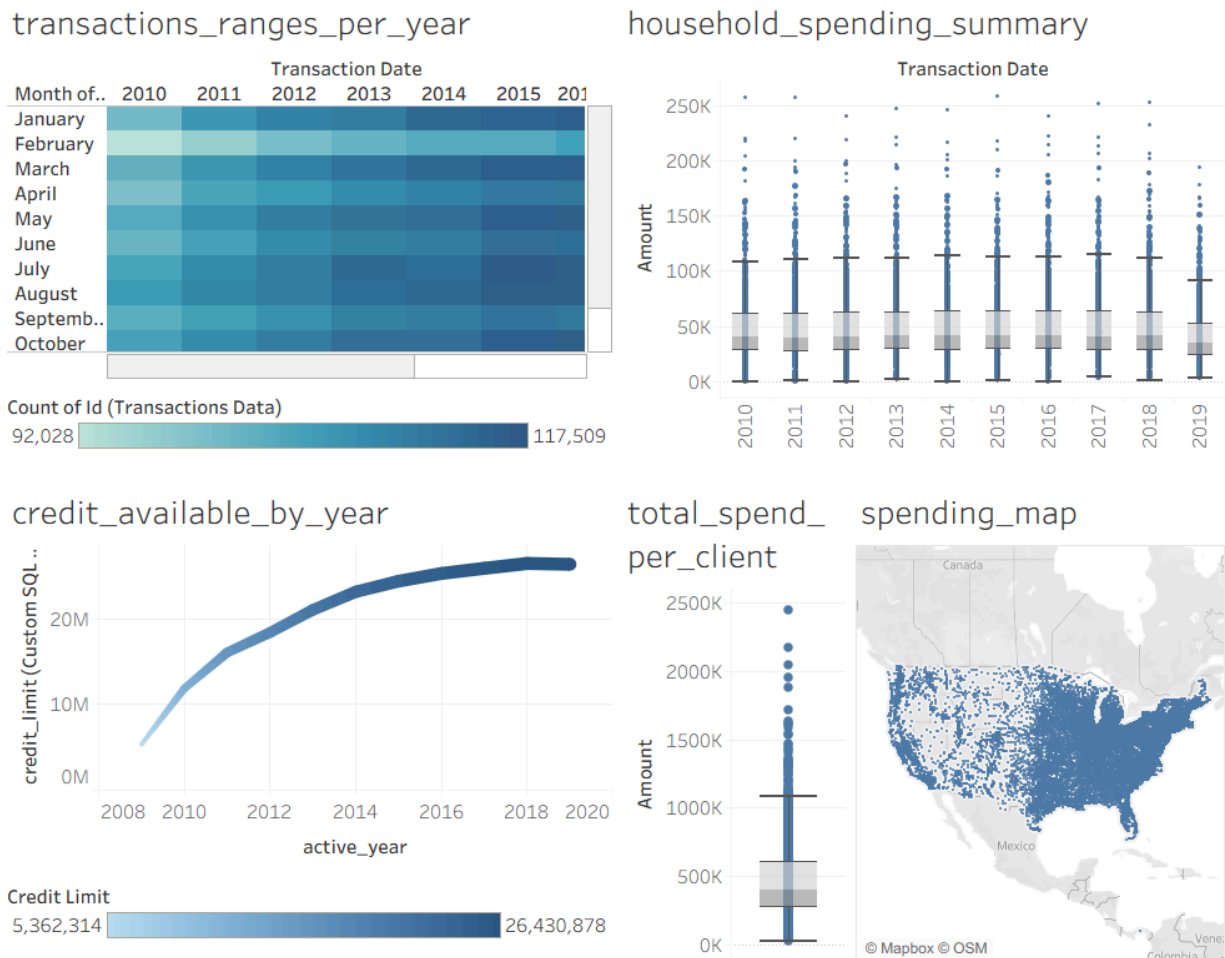


Figure 3. Exploratory data analysis of transaction patterns, household spending distributions, available credit over time, spending per client, and transaction geography.

Limitations:

Since the dataset does not contain comprehensive spending history on 18-24 year olds post 2009, finding a new dataset would give us further insights into their spending habits directly after the 2008 economic crisis. We are also limited by the number of clients this dataset uses (6146 Clients). More clients would give a better protection against sampling bias.

Future Development/Next Steps:

A natural next step is to apply the same data pipeline to a larger and more comprehensive dataset to reduce sampling bias and improve the stability of the trends we observed. We would also like to repeat the analysis on a more modern dataset that includes the post-COVID period, which would allow us to compare recovery patterns, shifts in online spending, and age-group behavior before and after the pandemic. We believe that the COVID shift to Online shopping would be very interesting to witness in the data.